# Toy pacifiers (toy-designated): HMTc Standard

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CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for toy pacifiers (toy-designated) in Children's Toys, Arts, and Crafts, evaluated on element MIGRATION from scraped-off toy material (EN 71-3 Category III, mg/kg of material): a migration ceiling, not a total-content ceiling. 8 of 11 panel analytes publish a numeric limit taken from the strictest applicable government maximum; the remainder are held under mandatory control or a substantiated not-material determination.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default; strictest applicable sovereign maximum in the row's native basis, disclosed read-across where the exact cell is unregulated; never occurrence-derived). Category 21 publishes on the element-migration basis only, pooling the EU, US and Mexican migration instruments, with each row pinned to the EN 71-3 material category or categories its scope actually spans.

## 2 | Master limit table

All values in ppb (µg/kg), as sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 23,000 | government maximum level |
| Cadmium (Cd) | Tier 1 | 17,000 | government maximum level |
| Total arsenic (tAs) | Tier 2 | 25,000 | government maximum level |

| | | | |
|---|---|---|---|
| Inorganic arsenic (iAs) | Tier 1 | n/m | not material in this matrix |
| Mercury, total (tHg) | Tier 1 | 60,000 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | n/m | not material in this matrix |
| Nickel (Ni) | Tier 2 | 930,000 | government maximum level |
| Tin (Sn) | Tier 2 | 180,000,000 | screening ceiling (European Union) |
| Aluminium (Al) | Tier 2 | 28,130,000 | government maximum level |
| Total chromium (Cr) | Tier 2 | 60,000 | screening ceiling (US FDA) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 53 | government maximum level |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 23,000 | government maximum level |
| Cadmium (Cd) | 17,000 | government maximum level |
| Total arsenic (tAs) | 25,000 | government maximum level |
| Inorganic arsenic (iAs) | n/m | No sovereign toy-safety instrument sets an inorganic-arsenic limit. Directive 2009/48/EC Annex II Part III point 13, EN 71-3, ISO 8124-3:2020 Table 1, ASTM F963-17 4.3.5 and NOM-252-SSA1-2011 Table 1 all regulate TOTAL arsenic migration from toy material; inorganic arsenic is a dietary speciation with no article-migration counterpart. Arsenic in a toy-designated pacifier teat, shield and clip is bounded by the published total-arsenic migration ceiling by identity (iAs <= tAs), so a separate inorganic-arsenic ceiling would add no control. |

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| | | |
|---|---|---|
| Mercury, total (tHg) | 60,000 | government maximum level |
| Methylmercury (MeHg) | n/m | Methylmercury is produced by microbial methylation in aquatic systems and reaches humans by diet. a toy-designated pacifier teat, shield and clip carries no methylation pathway and no dietary route, and no sovereign toy instrument sets a methylmercury limit: every toy mercury limit (2009/48/EC point 13, ASTM F963-17 4.3.5, ISO 8124-3:2020, NOM-252-SSA1-2011, Canada SOR/2011-17 and SOR/2016-193) is on total mercury. Any organomercury fraction is bounded by the published total-mercury migration ceiling by identity (MeHg <= tHg). |
| Nickel (Ni) | 930,000 | government maximum level |
| Tin (Sn) | 180,000,000 | government maximum level |
| Aluminium (Al) | 28,130,000 | government maximum level |
| Total chromium (Cr) | 60,000 | government maximum level |
| Hexavalent chromium (Cr-VI) | 53 | government maximum level |

## 4 | Cite this standard

Pendergrass, K. (2026). *Toy pacifiers (toy-designated): HMTc standard*. Institute of Contaminant Standards.

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### STATUS AND LICENCE

A normative certification standard, editorially reviewed and version-controlled in-house; every value traces to a named sovereign maximum level, disclosed in the derivation. © 2026 Institute of Contaminant Standards / Heavy Metal Certified. All rights reserved.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the briefing page at https://heavymetalcertified.com/standards/toy-pacifiers. Primary legal instruments and expert-body assessments are

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This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/toy-pacifiers.

named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

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